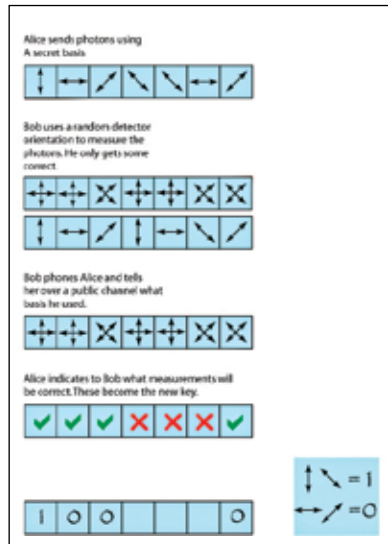


really smart)? Check out Shor's algorithm for prime factoring. Really, please do. There are few mathematical formulae in the world that can bring it to its knees the moment one knows how to use them. No kidding.

Quantum Cryptography

We might not have built a quantum computer, but the field has seen incredible advancements, and today you can possibly use quantum techniques in your friendly neighbourhood ciphers.

Let us take, for example, something called the BB84 cryptography protocol. The super stripped-down idea is that if Alice wants to send data to Bob (because apparently, when it comes to a quantum situation needing 2 people, it is always Alice and Bob), what would be done is that Alice would encode her data in qubits (something like photons - the light particles), and send them to Bob in some arbitrary bases. Now, Bob would receive the inflow in some arbitrary bases of his choice. His results would be phased by the difference of bases chosen by him and those chosen by Alice. This problem, though can easily be corrected by a simple communication to clear the air on this topic. Oh, and forget about eavesdropping, because if a third listener tried to copy the data with them, it would not be allowed. The network won't stop him, and neither would Bob or Alice. That's because they don't need to - the Universe is on it already. It is impossible to copy quantum bits exactly.



The process explained graphically

Post-Quantum Cryptography

There are many, many allegations that you might throw at mankind, and most of them would probably be true. But one thing that you can never say that humans lack is vision. Yes, today's attempts at Quantum Com-